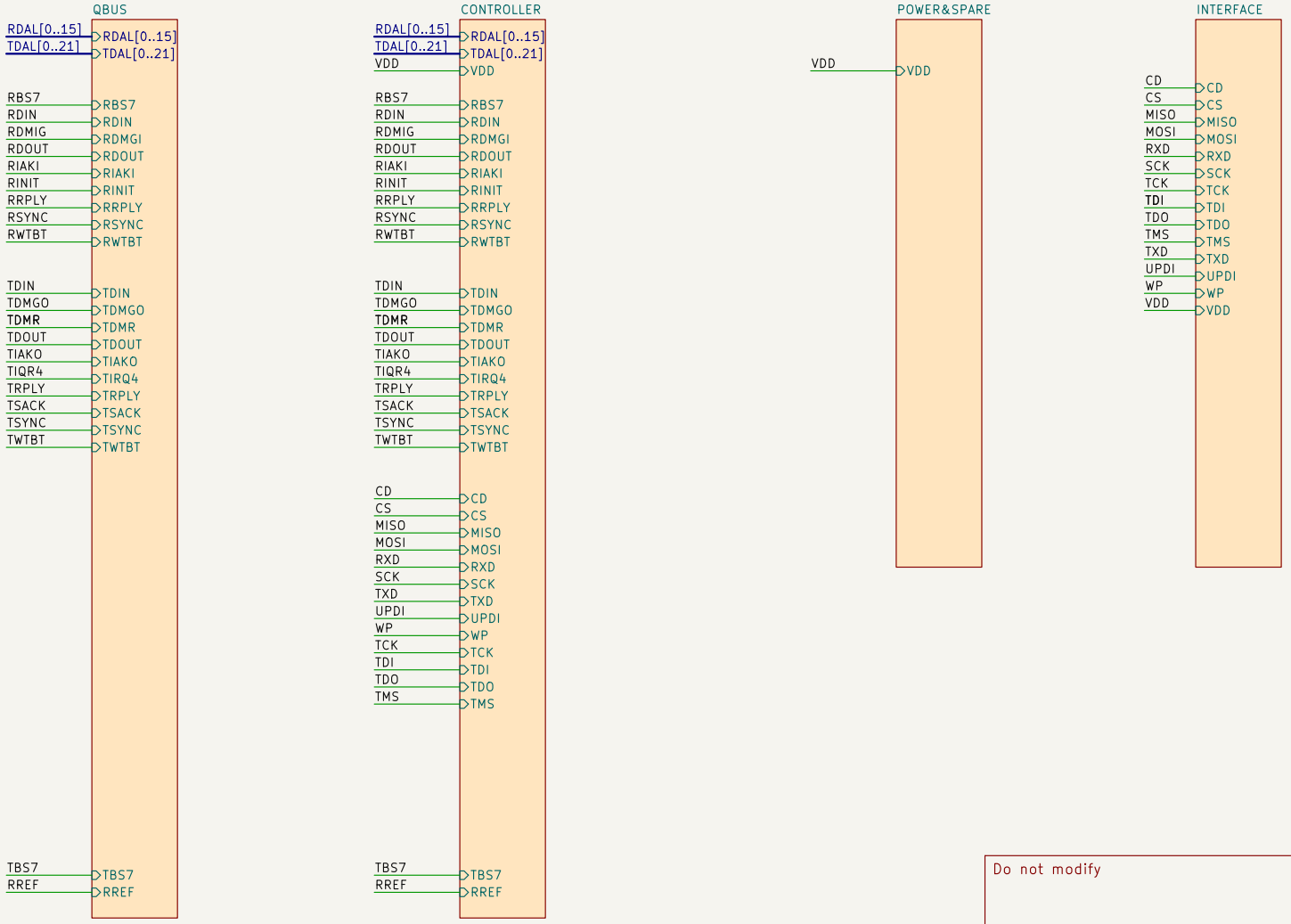
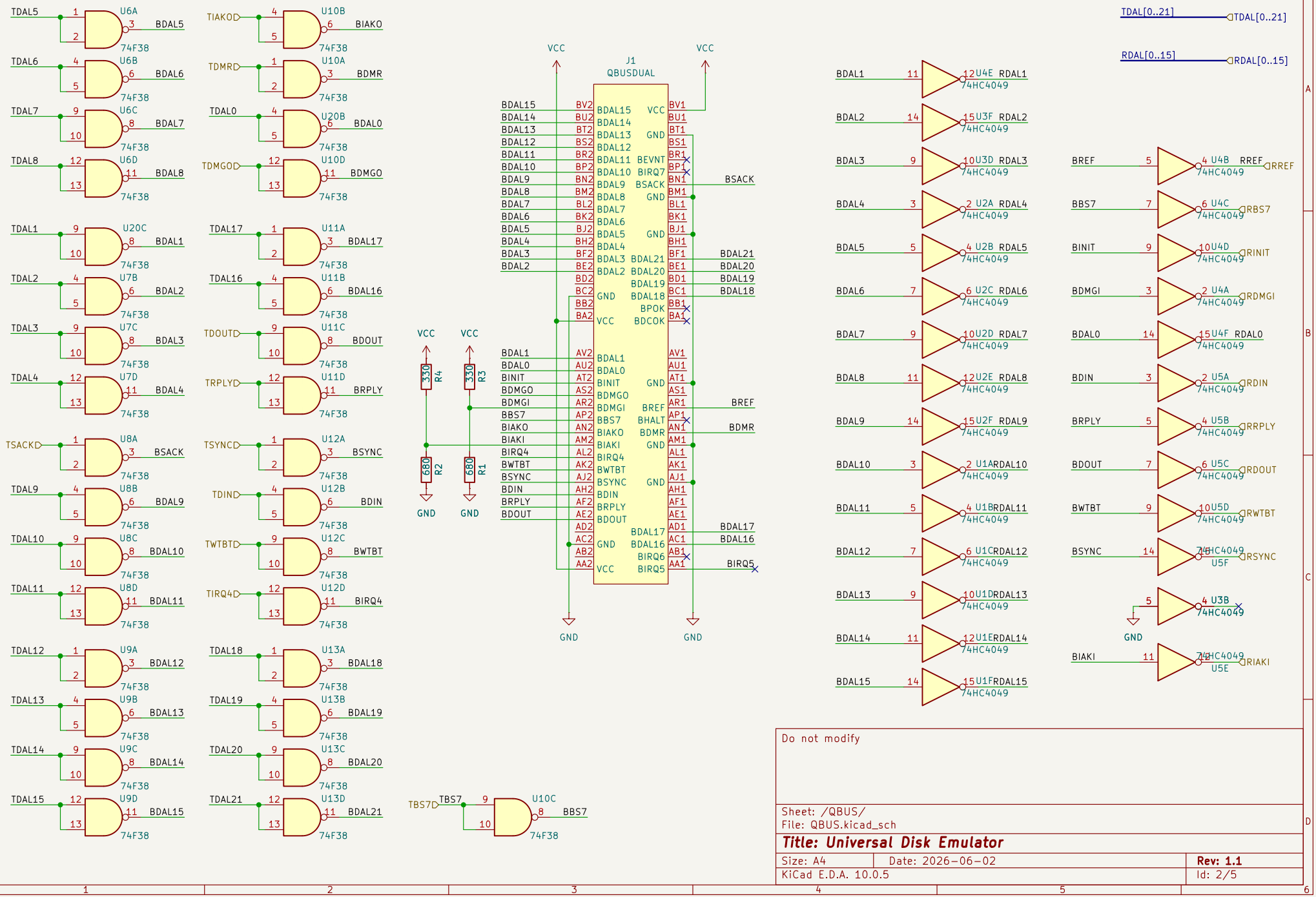


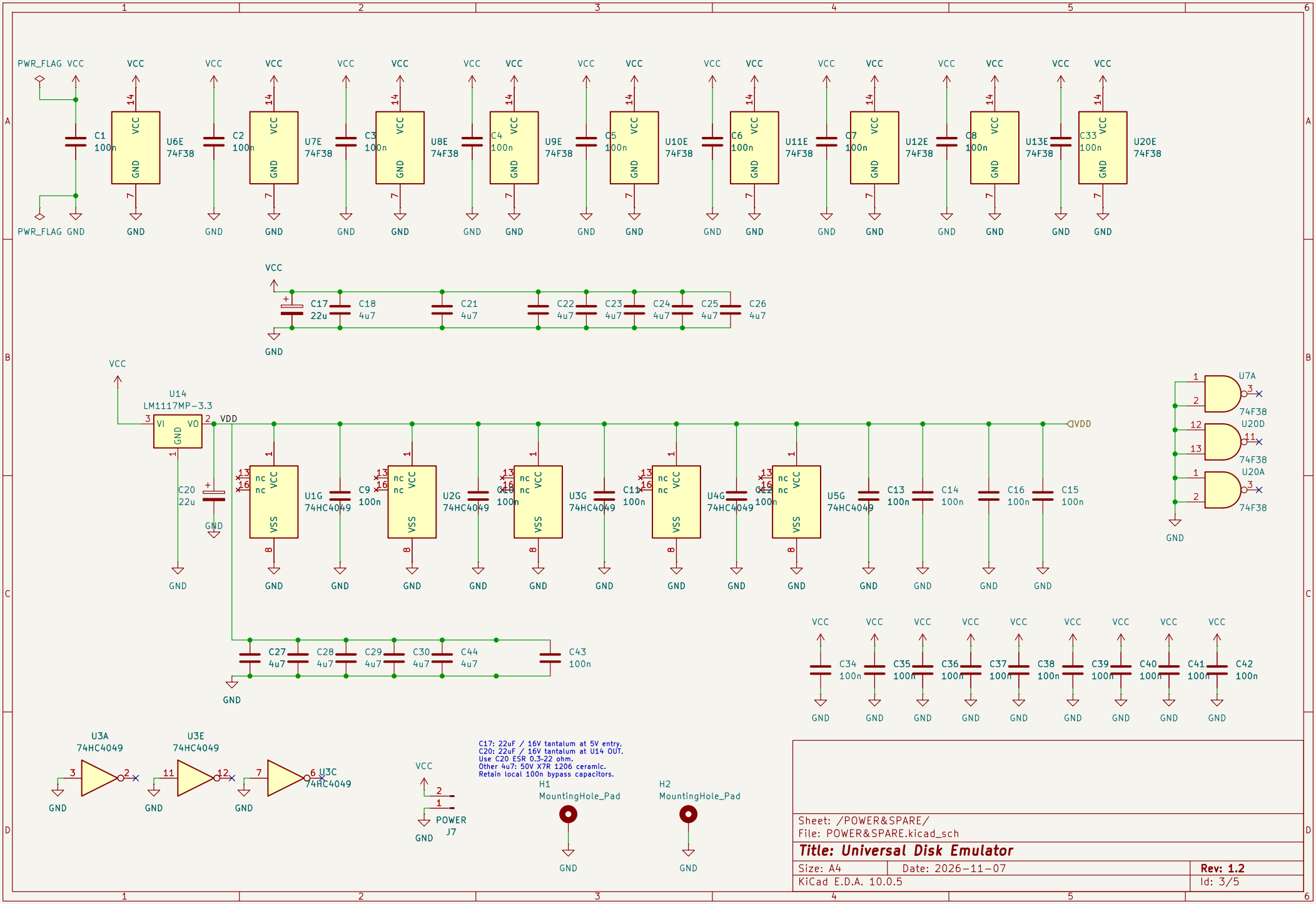
Do not modify

Sheet: /
File: DISKEmu-v5-1.kicad_sch

Title: Universal Disk Emulator

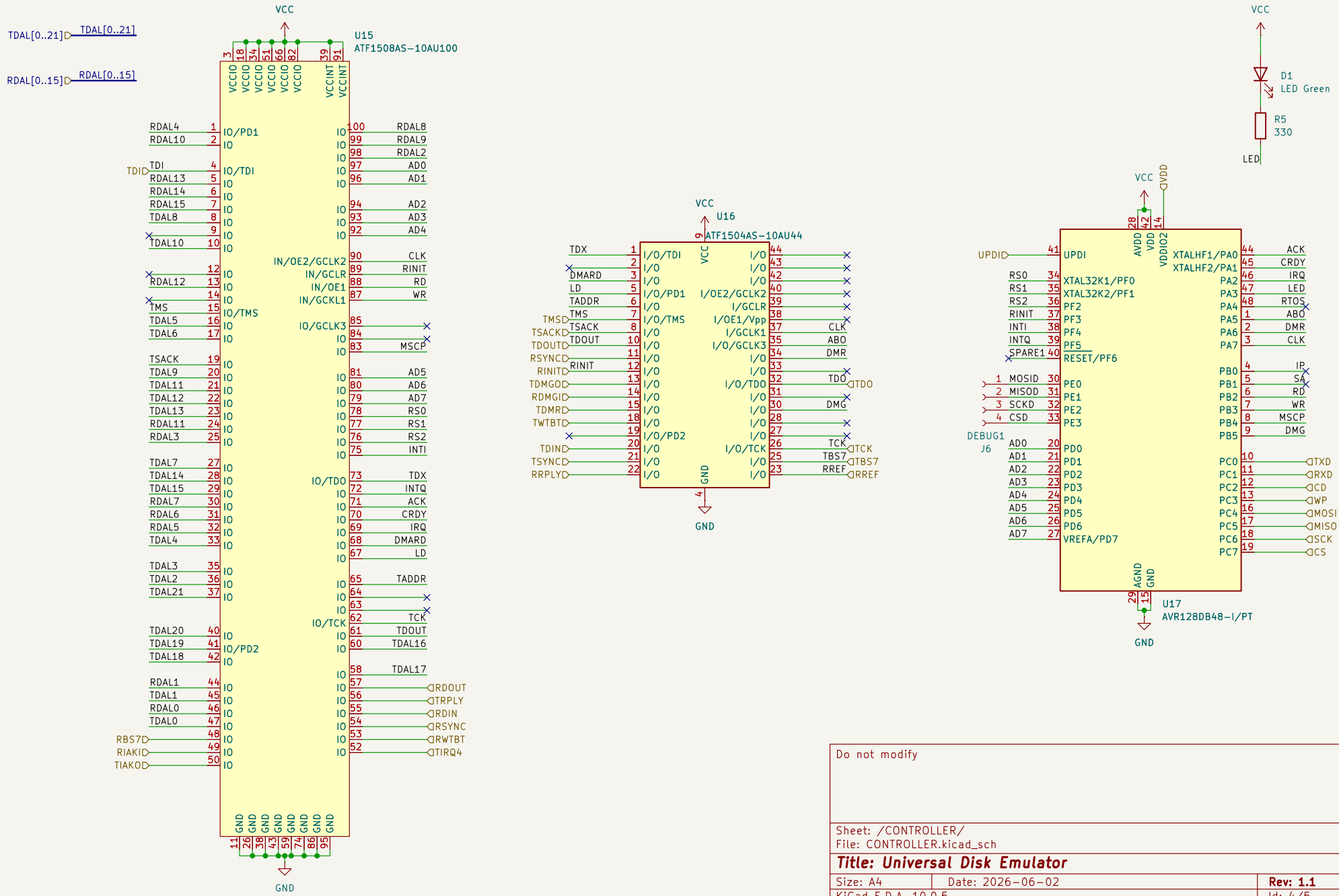
Size: A4	Date: 2026-06-02	**Rev: 1.1**
KiCad E.D.A. 10.0.5		Id: 1/5


TDAL5
U6A
74F38
BDAL5
TDAL6
U6B
BDAL6
TDAL7
U6C
BDAL7
TDAL8
U6D
BDAL8
TIAKO
U10B
BIAKO
TDMR
U10A
BDMR
TDAL0
U20B
BDAL0
TDMGO
U10D
BDMGO
TDAL1
U20C
BDAL1
TDAL2
U7B
BDAL2
TDAL3
U7C
BDAL3
TDAL4
U7D
BDAL4
TDAL17
U11A
BDAL17
TDAL16
U11B
BDAL16
TDOUT
U11C
BDOUT
TRPLY
U11D
BRPLY
TSACK
U8A
BSACK
TDAL9
U8B
BDAL9
TDAL10
U8C
BDAL10
TDAL11
U8D
BDAL11
TSYNC
U12A
BSYNC
TDIN
U12B
BDIN
TWTBT
U12C
BWTBT
TIRQ4
U12D
BIRQ4
TDAL12
U9A
BDAL12
TDAL13
U9B
BDAL13
TDAL14
U9C
BDAL14
TDAL15
U9D
BDAL15
TDAL18
U13A
BDAL18
TDAL19
U13B
BDAL19
TDAL20
U13C
BDAL20
TDAL21
U13D
BDAL21
VCC
J1
QBUSDUAL
GND
330
R4
R3
680
R2
R1
TBS7
U10C
BBS7
TDAL[0..21]
RDAL[0..15]
BDAL1
U4E
RDAL1
74HC4049
BDAL2
U3F
RDAL2
BDAL3
U3D
RDAL3
BDAL4
U2A
RDAL4
BDAL5
U2B
RDAL5
BDAL6
U2C
RDAL6
BDAL7
U2D
RDAL7
BDAL8
U2E
RDAL8
BDAL9
U2F
RDAL9
BDAL10
U1A
RDAL10
BDAL11
U1B
RDAL11
BDAL12
U1C
RDAL12
BDAL13
U1D
RDAL13
BDAL14
U1E
RDAL14
BDAL15
U1F
RDAL15
BREF
U4B
RREF
BBS7
U4C
RBS7
BINIT
U4D
RINIT
BDMGI
U4A
RDMGI
BDAL0
U4F
RDAL0
BDIN
U5A
RDIN
BRPLY
U5B
RRPLY
BDOUT
U5C
RDOUT
BWTBT
U5D
RWTBT
BSYNC
U5F
RSYNC
U3B
BIAKI
U5E
RIAKI
Do not modify
Sheet: /QBUS/
File: QBUS.kicad_sch
Title: Universal Disk Emulator
Size: A4
Date: 2026-06-02
Rev: 1.1
KiCad E.D.A. 10.0.5
Id: 2/5



Sheet: /POWER&SPARE/	
File: POWER&SPARE.kicad_sch	
Title: Universal Disk Emulator	
Size: A4	Date: 2026-11-07
KiCad E.D.A. 10.0.5	Id: 3/5

Rev: 1.2


TDAL[0..21]
RDAL[0..15]
U15
ATF1508AS-10AU100
VCC
GND
U16
ATF1504AS-10AU44
U17
AVR128DB48-I/PT
D1
LED Green
R5
330
LED
DEBUG1
J6
Do not modify
Sheet: /CONTROLLER/
File: CONTROLLER.kicad_sch
Title: Universal Disk Emulator
Size: A4
Date: 2026-06-02
Rev: 1.1
KiCad E.D.A. 10.0.5
Id: 4/5

USB / PROGRAMMING — FT2232H

Self-powered from board 3.3 V. USB and board share ground.
USB VBUS is sensed only; it does not power the board.

J2
TYPE-C-31-M-12

VBUS A4 → USB_VBUS
CC1 A5 → USB_CC1
CC2 B5 → USB_CC2
D- A7 / B7 → USB_D-
D+ A6 / B6 → USB_D+
SBU1 A8
SBU2 B8
SHIELD → SH → GND
GND A1 / B12

R6 5k1 USB_CC1 → GND
R7 5k1 USB_CC2 → GND

C31 100n GND

Y1
12MHz 20pF
XTAL_IN1 — 3 — XTAL_OUT
2 → GND

XTAL_IN → C46 10n C0G → GND
XTAL_OUT → C47 10n C0G → GND

U22
74LVC1G07
VDD → 5
USB_VBUS 2 → GND 3
4 → USB_RESET_N

USB_VBUS → R12 2k → GND
C48 100n VDD → GND
R13 2k VDD → USB_RESET_N

U22: VBUS absent holds U18 in reset.
5.5 V tolerant input: Ioff prevents back-powering.
Keep USB D+/D- short, matched and 90 ohm differential.

U18
FT2232HL

VDD 50 → VREGIN
VREGOUT 49 → USB_1V8
USB_PHY_3V3 → VPHY 4
USB_PLL_3V3 → VPLL 9
USB_1V8 → VCORE 12
USB_1V8 → VCORE 37
USB_1V8 → VCORE 64
VDD → VCCIO 20
VDD → VCCIO 31
VDD → VCCIO 42
VDD → VCCIO 56

USB_D- 7 → DM
USB_D+ 8 → DP
USB_REF 6 → REF
USB_RESET_N 44 → RESET

EE_CS 63 → EECS
EE_CLK 62 → EECLK
EE_DATA 61 → EEDATA

XTAL_IN 2 → OSCI
XTAL_OUT 3 → OSCO
GND 13 → TEST

ADBUS0 16 → FT_TCK
ADBUS1 17 → FT_TDI
ADBUS2 18 → TDO
ADBUS3 19 → FT_TMS
ADBUS4 21
ADBUS5 22
ADBUS6 23
ADBUS7 24

ACBUS0 26
ACBUS1 27
ACBUS2 28
ACBUS3 29
ACBUS4 30
ACBUS5 32
ACBUS6 33
ACBUS7 34

BDBUS0 38 → FT_TX
BDBUS1 39 → FT_RX
BDBUS2 40
BDBUS3 41
BDBUS4 43
BDBUS5 45
BDBUS6 46
BDBUS7 47

BCBUS0 48
BCBUS1 52
BCBUS2 53
BCBUS3 54
BCBUS4 55
BCBUS5 57
BCBUS6 58
BCBUS7 59

PWREN 60
SUSPEND 36

AGND 10 → GND
GND 1 / 14 / 15 / 25 / 35 / 41 / 51

CHANNEL A — CPLD JTAG

JP2
ONBOARD JTAG ENABLE
FT_TCK 1 — 2 → TCK
FT_TDI 3 — 4 → TDI
FT_TMS 5 — 6 → TMS

Fit 1–2, 3–4, 5–6 for onboard USB JTAG.
Remove all three before using external J3.
A0=TCK, A1=TDI, A2=TDO, A3=TMS.
Chain: U15 ATF1508 -> U16 ATF1504.

J3
CPLD JTAG
VCC → VREF 4
VCC 7
SRST 6
TRST 8
TCK 1 → TCK
TMS 5 → TMS
TDO 3 → TDO
TDI 9 → TDI
GND 2
GND

R14 2k VDD → TMS
R15 2k GND → TCK

CHANNEL B — CONSOLE / AVR UPDI

JP1
CONSOLE / UPDI
RXD 1 — 2 → TXD
FT_TX 3 — 4 → FT_RX
UPDI_TX 5 — 6 → UPDI

CONSOLE: shunt 1–3 and 2–4.
UPDI: shunt 3–5 and 4–6.
Move both shunts together: do not bridge 1–2 / 3–4 / 5–6.
B0=TX, B1=RX: UART hardware flow control Off.

U19
74LVC1G07
VDD → 5
UPDI_TX 2 → GND 3
4 → UPDI_DD
1 → GND

UPDI_DD — 330Ω 1% — UPDI

R17 2k VCC → UPDI
R18 2k VDD → UPDI_DD
C49 100n VDD → GND

J4
AVR UPDI
VCC → 2
DATA 1 → UPDI
GND 6
GND

UPDI high level = AVR 5 V supply.
Open-drain TX releases the line for AVR replies.
R17 = 2k2 for faster UPDI rising edges.
Keep traces short; check edges at selected baud rate.
Remove JP1 shunts before using external J4.

USB CONFIGURATION EEPROM

U21
93LC46B-/SN (x16)
VDD → 8
EE_CS 1 → CS
SCLK 2 → EE_CLK
DI 3 → EE_DATA
DO 4 → EE_Q
GND 5
GND

EE_Q — 2k2R19 — EE_DATA

R20 2k VDD → EE_DATA
R21 2k GND → EE_CS
C50 100n VDD → GND

Program EEPROM over USB with FT_PROG:
Self-powered; channel A 245X/MPSSE, B UART/VCP.
Set unique serial number. Disable suspend pull-downs.
VBUS detection uses RESET, not the channel-B suspend option.

BOARD SIGNALS

VDD → VDD
TCK → TCK
TDI → TDI
TDO → TDO
TMS → TMS
UPDI → UPDI
TXD → TXD

RXD → RXD
CS → CS
MOSI → MOSI
MISO → MISO
SCK → SCK
CD → CD
WP → WP

GND → GND
VCC → VCC

FULL-SIZE SD CARD — SHOU HAN C7529393

J5
SD-CARD 11P
CS 1 → CD/DAT3
MOSI 2 → CMD/DI
GND 3 → VSS1
VDD 4 → VDD
SCK 5 → CLK
GND 6 → VSS2
MISO 7 → DAT0/DO
SD_DAT1 8 → DAT1
SD_DAT2 9 → DAT2
CD 11 → CD
WP 10 → WP
SHELL/COMMON → SH → GND

R8 2k SD_DAT1 → VDD
R9 2k SD_DAT2 → VDD
R10 2k MOSI → VDD

FT2232H POWER DECOUPLING

C19 100n USB_1V8 → GND
C32 100n USB_1V8 → GND
C45 100n VDD → GND
C51 100n USB_1V8 → GND
C52 100n USB_1V8 → GND
C53 100n VDD → GND
C54 100n VDD → GND
C55 100n VDD → GND
C56 100n VDD → GND
C57 100n VDD → GND
C58 100n VDD → GND
C59 100n USB_PHY_3V3 → GND

FB1 600R@100MHz VDD → USB_PHY_3V3
FB2 600R@100MHz VDD → USB_PLL_3V3

C40 1u USB_PHY_3V3 → GND
C41 1u USB_PLL_3V3 → GND

100n at every VCCIO/VCORE/VREGIN/VPHY/VPLL pin.
C19: effective capacitance >=3.3uF at 1.8 V.
All 4u7 ceramics: 50 V X7R, 1206; verify DC-bias curves.
Do not connect USB VBUS to board VCC or VDD.

PWR_FLAG → USB_VBUS
PWR_FLAG → USB_PHY_3V3
PWR_FLAG → USB_PLL_3V3

LAYOUT: retain short capacitor returns; check U14 thermal margin with added FT2232H load. External JTAG and UPDI headers retained.
J3/J4 reference is board 5 V: do not feed USB power into either header.

Sheet: /INTERFACE/	
File: INTERFACE.kicad_sch	
Title: Universal Disk Emulator — USB programming / SD	
Size: A2	Date: 2026-11-07
KiCad E.D.A. 10.0.5	Rev: 1.2
	Id: 5/5